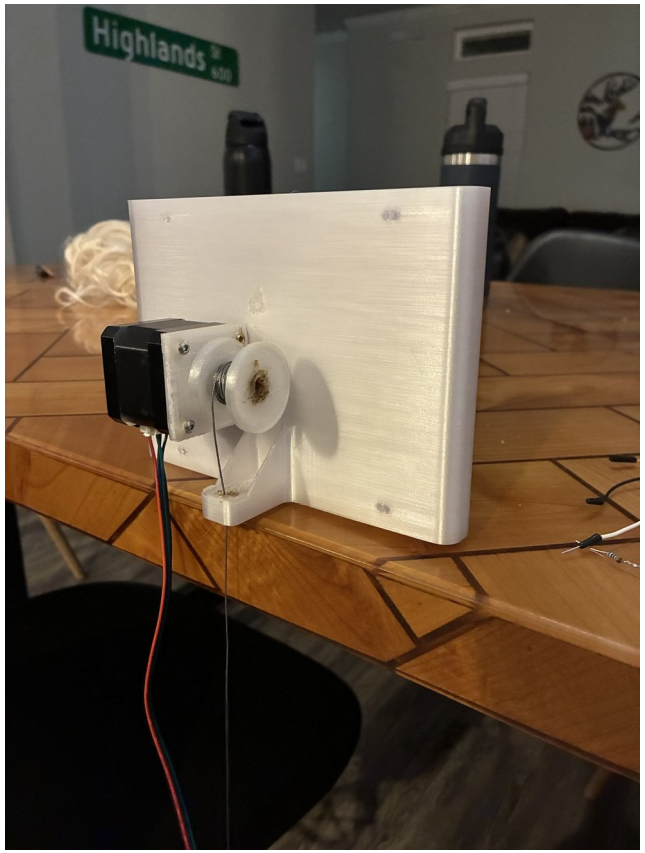
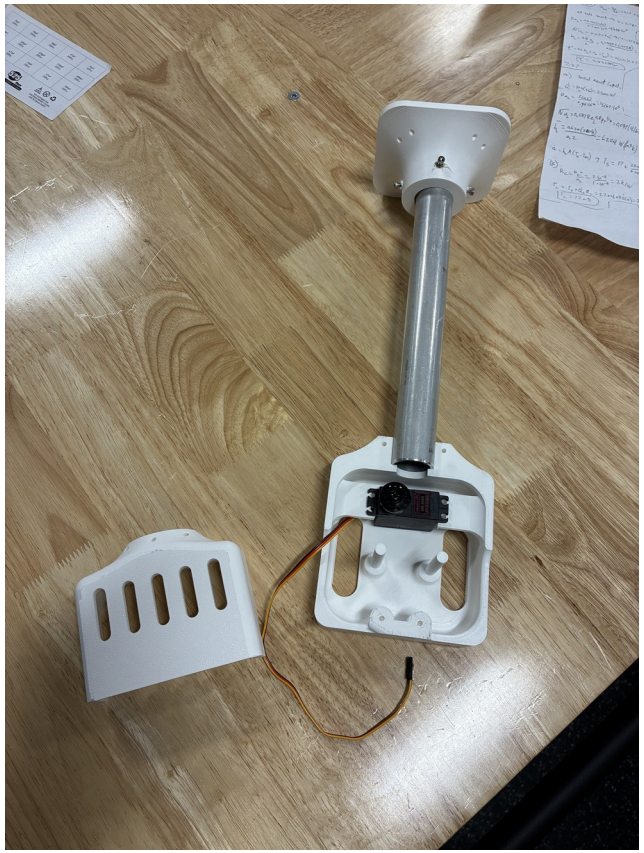
# RoomCleaner

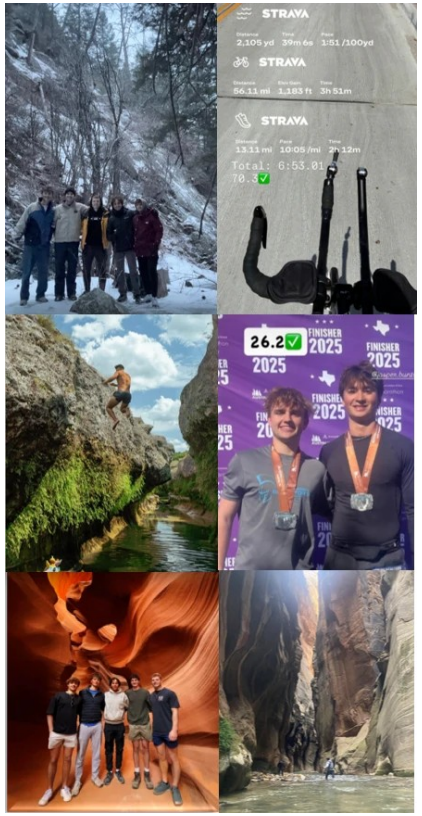
A cable-driven robot that picks laundry up off the floor

Jasper Buntinx
Mechanical Engineering, Texas A&M University
October 27, 2026

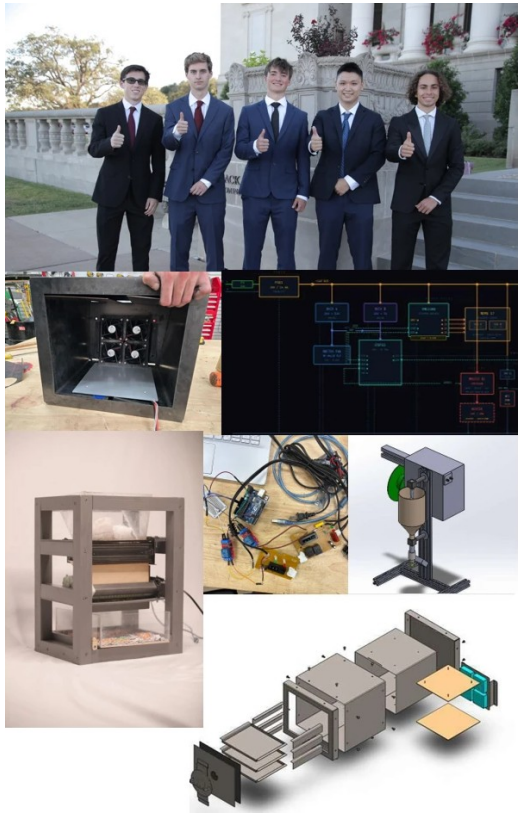


# About Me




Outdoors and endurance



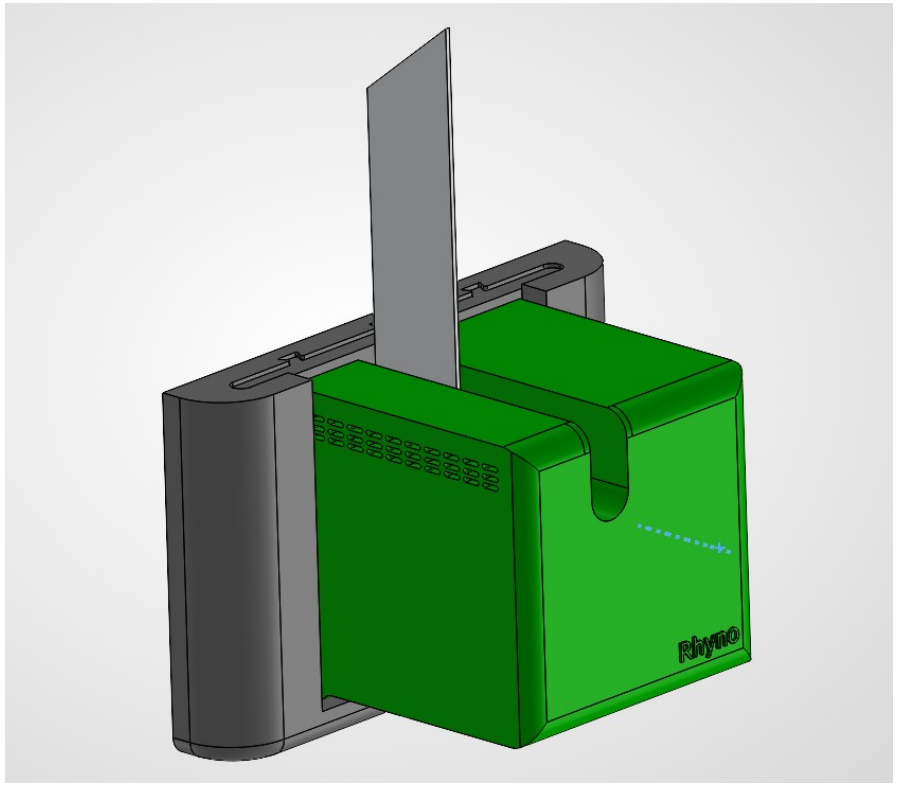
Engineering projects




Community

- Mechanical Engineering, Texas A&M University, class of 2027
- Built EcoFil, a filament recycler with a shredder, dryer and extruder
- Finished an Ironman 70.3 in 6:53 and a marathon in 2025
- Hiking, canyons and cliff jumping
- Led obstacle assembly, repairs and transport for the Gladiator Dash, which raised $185K

# Project Goal



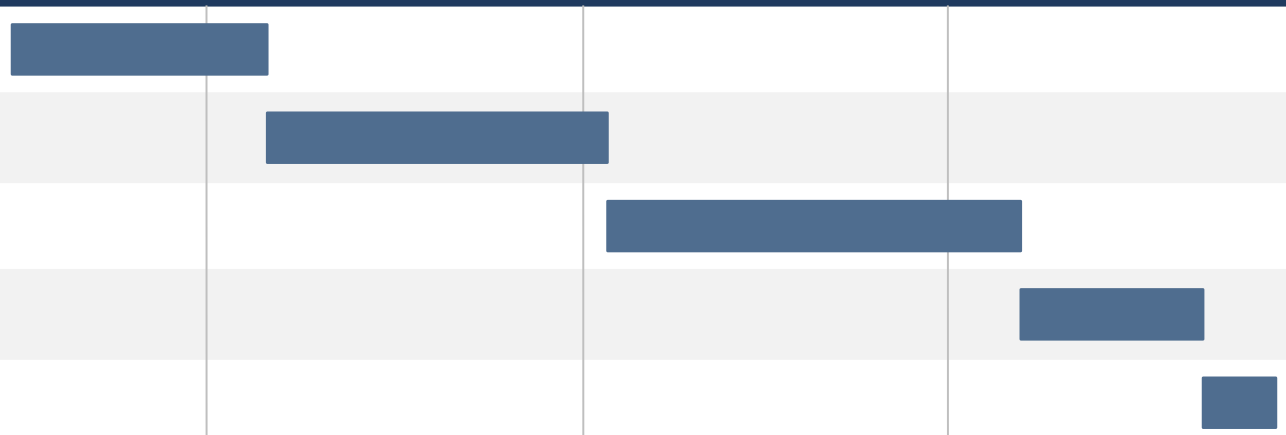



- Pick up laundry anywhere in an 11 × 9 ft room and drop it in a hamper
- Design load: jeans, 0.9 kg, at 2 m/s²; first test with 1 lb
- 3D-printed PETG (polyethylene terephthalate glycol) on a hobby budget
- Success: one garment to the hamper with no human input

Final design (SolidWorks): wall station and claw

# Timeline and Scope

| Phase | Dates | July | August | September | October |
|---|---|---|---|---|---|
| 1. Concept | Jul 16 – Aug 5 | | | | |
| 2. Generation 1 | Aug 6 – Sep 2 | | | | |
| 3. Redesign | Sep 3 – Oct 6 | | | | |
| 4. Final design | Oct 7 – Oct 21 | | | | |
| 5. Build and test | Oct 22 – Oct 27 | | | | |



- Solo project: requirements, calculations, all CAD (computer-aided design), printing, machining, wiring and *[software parts you wrote]*
- Purchased: stepper motors, drivers, controller boards, servo, camera, battery and encoder boards
- Garment detection uses YOLO-World, an open-source model

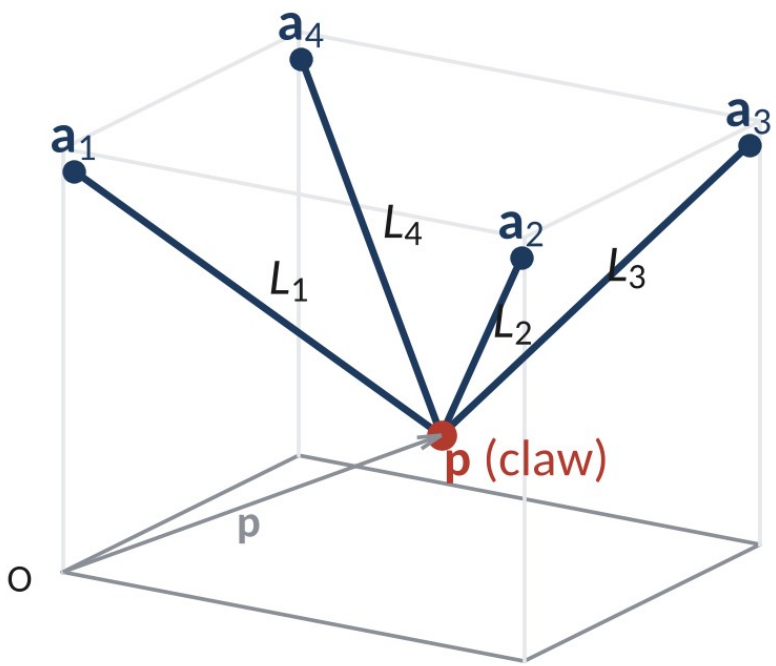
Phase 1

# Concept

July 16 – August 5

Choose the architecture and set the limits

# Why a Cable Robot




- Four ceiling winches hold one claw: the whole floor is reachable and nothing stands on it
- The four cable lengths set the claw position
- A laptop plans every move; a phone app starts, pauses and approves it

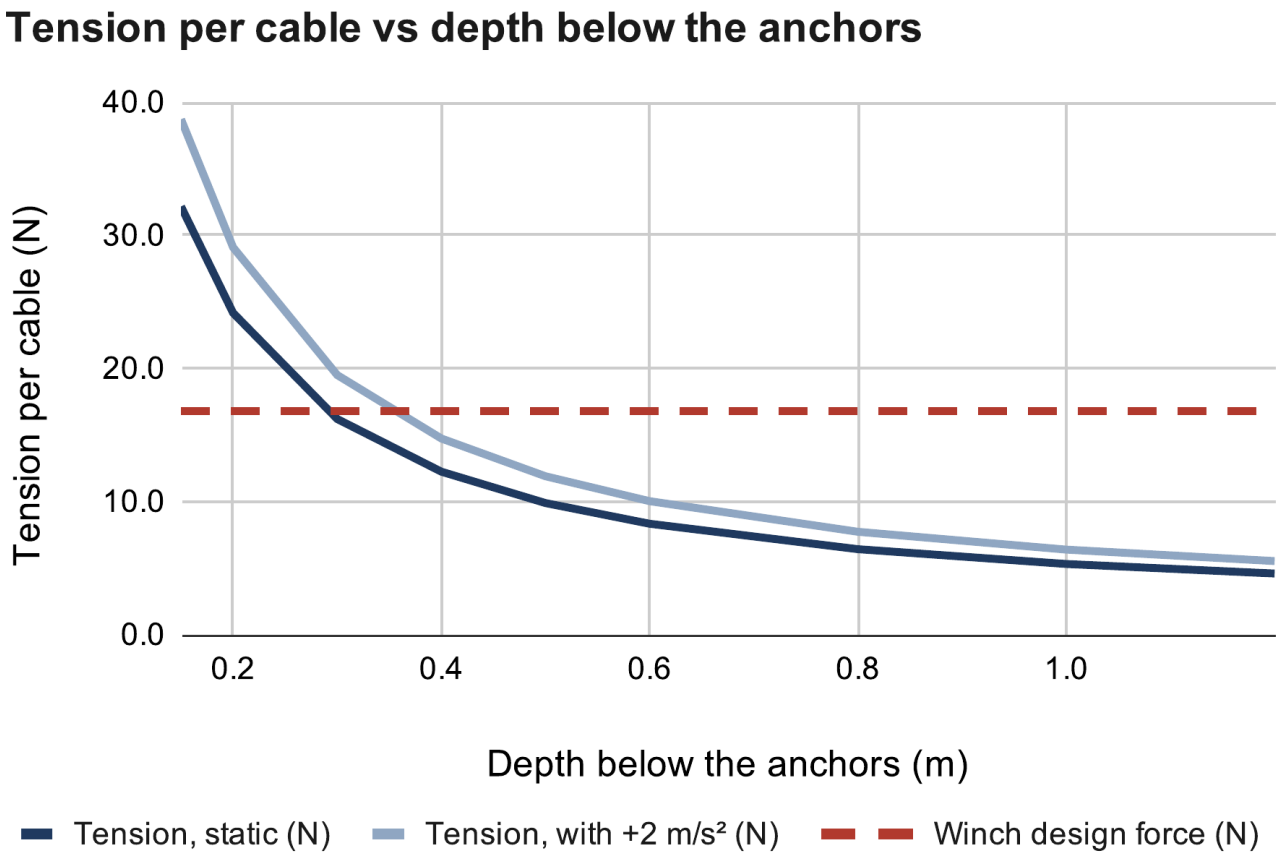
Cables run from the ceiling anchors $a_1$ to $a_4$ to the claw at p

# Design Trade-offs

| Design change | Benefit | Cost | Chosen setting |
|---|---|---|---|
| **Higher motor current** | 41 % more pull at 1.5 A | Heat rises with current squared | 1.5 A, vented shell and heat fin |
| **Stronger claw** | Gear safety factor 0.4 to 2.0 | More claw mass, less payload | Module 1.5 gears |
| **Larger room** | More floor covered | Longer wires, more voltage drop | Only 12 V power runs to stations |
| **Closer to the ceiling** | More working volume | Cable tension 12 N to 32 N | Stay 0.4 m below the anchors |
| **Smaller spool radius** | More pull per amp | Slower line speed | 15 mm core: 25–34 N pull |
| **Larger safety margins** | No skipped steps or wall contact | Less usable force and floor | Force ÷ 1.5; 0.2 m from walls |

- A heavier claw needs more pull, more pull needs more current, and heat rises with the square of the current

# Cable Tension Limit




Source: RoomCleaner engineering calculations, “Cable tension” (calculated)

- Tension rises sharply near the ceiling: 32 N at 0.15 m, 8.3 N at 0.6 m
- Operating limit set 0.4 m below the anchors
- Covers 2 m/s² acceleration within the 16.8 N winch limit

**Tension per cable at 0.4 m**

$T = W / (4 \sin \varphi)$

$= 8.87\ \text{N} / (4 \times 0.181) = \mathbf{12.2\ N}$

With 2 m/s²: **14.7 N** < 16.8 N

Phase 2

# Generation 1

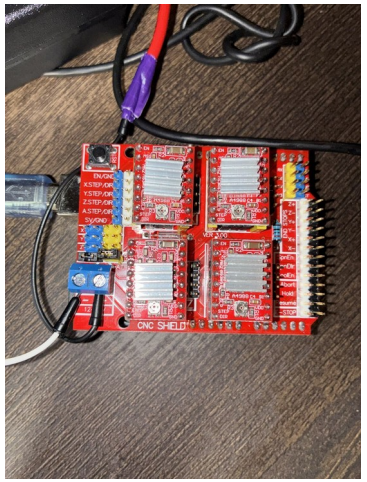
August 6 – September 2

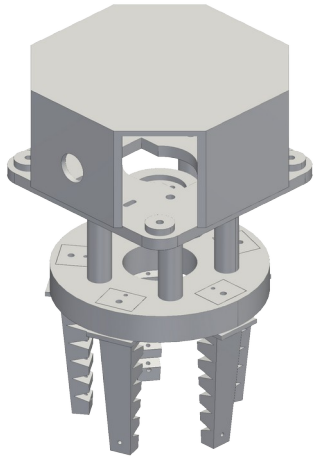
Prove the idea on the bench

# Generation 1 Build





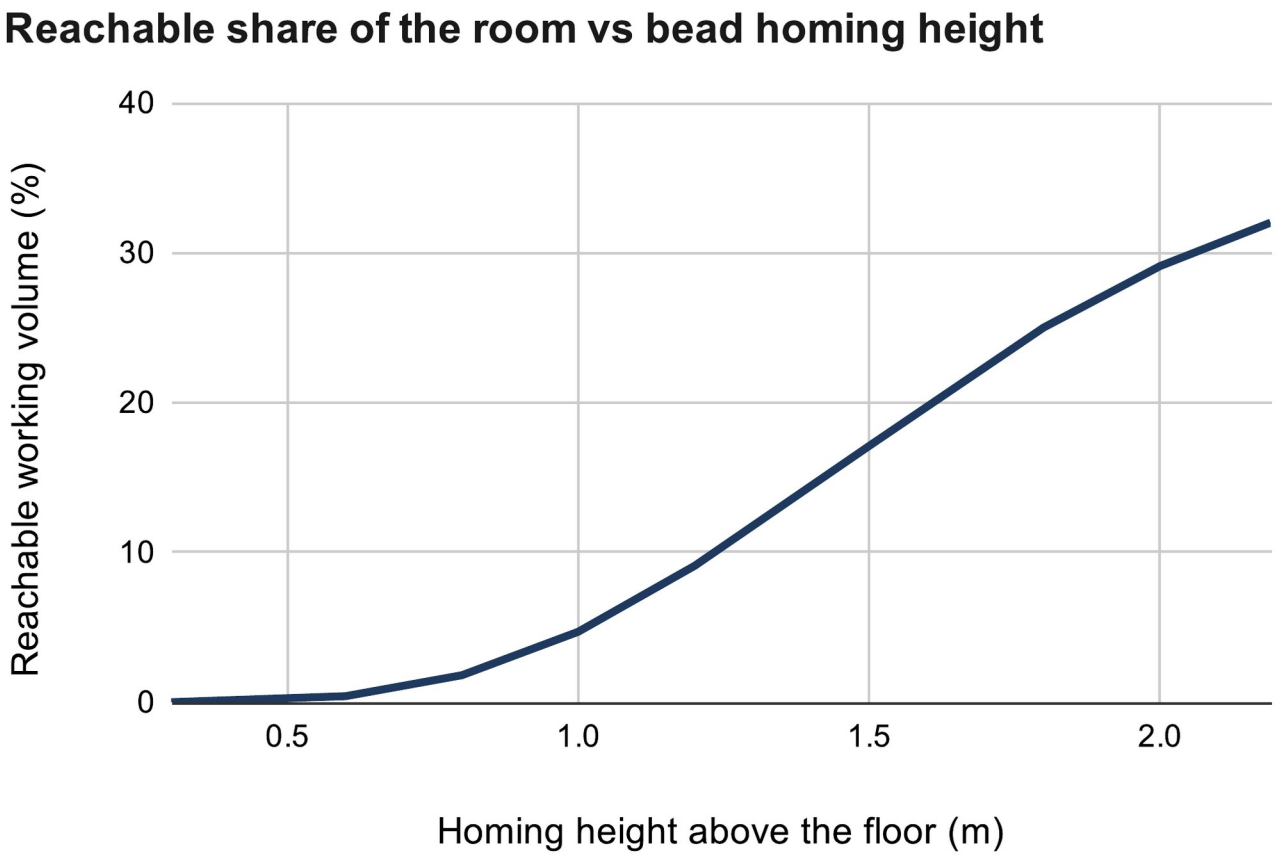




Bench lift test, corner plate, central driver board, finger claw (CAD)

- One-piece corner plates: motor, spool, pulley and home switch
- One central board drove all four motors
- Five flexible TPU (thermoplastic polyurethane) fingers on the claw
- A station lifted a test weight on the bench *[weight]*

# What Generation 1 Taught Me




- Bead homing left only 5–29 % of the room reachable
- Adding a sensor at each winch meant 10 wires per corner
- Finger claw could not pinch fabric and was 4.6 times over its mass budget

Source: RoomCleaner engineering calculations, “Bead homing” (calculated)

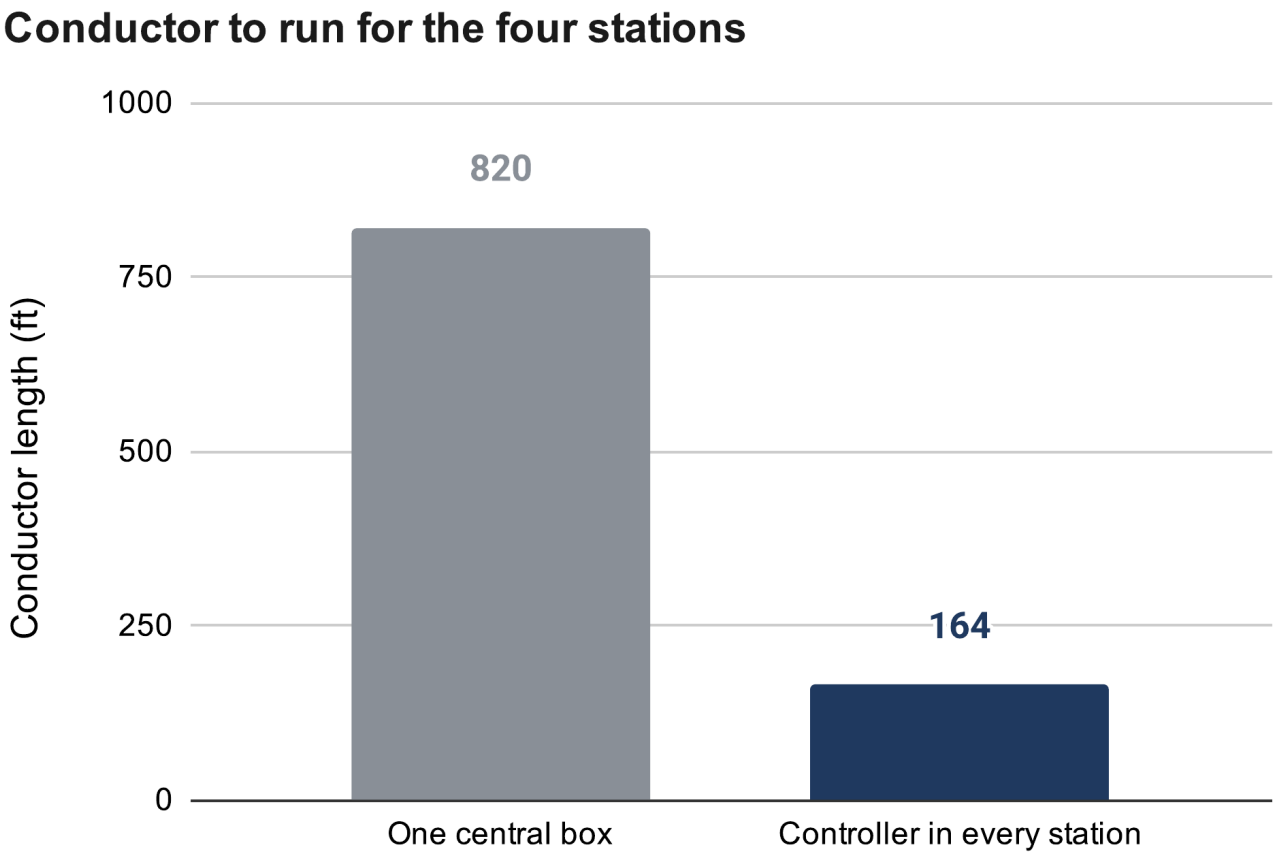
Phase 3

# Redesign

September 3 – October 6

Fix what generation 1 exposed

# One Controller per Winch




Source: RoomCleaner engineering calculations, “Wiring” (calculated)

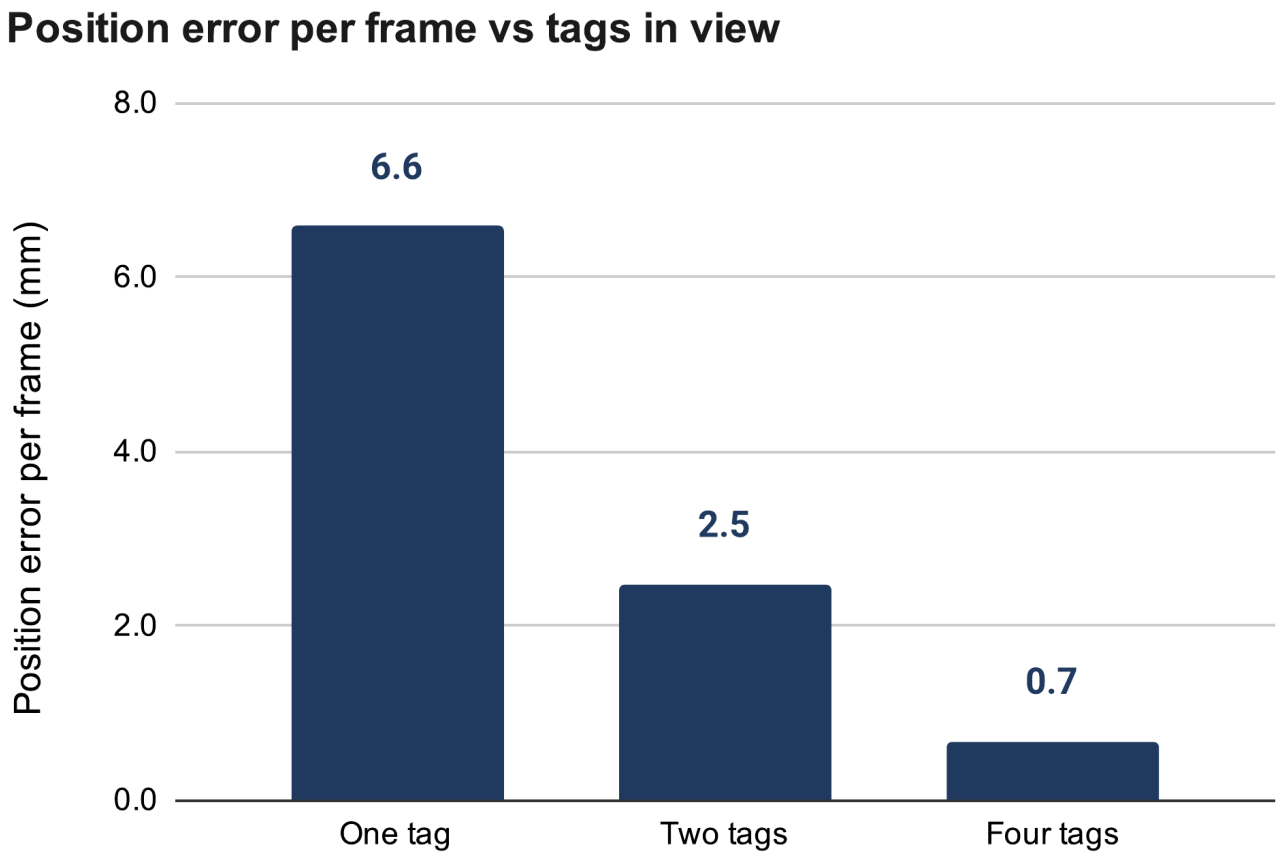
- A controller in each station instead of one central board
- Only a 12 V pair runs to each station: five times less wire
- Cost: synchronization moves into software

**Wire to the four stations**

Central board: 10 × 82 ft = **820 ft**

Per station: 2 × 82 ft = **164 ft**

# Camera Calibration Replaces Homing

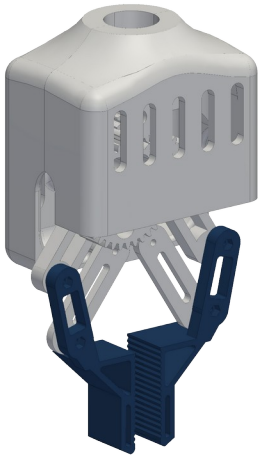



- The claw camera reads a printed floor board from several poses
- A fit finds the anchor positions and cable lengths: no tape measure
- Position error: 6.6 mm with one tag in view, 0.7 mm with four
- Whole room reachable; the beads stay as end stops

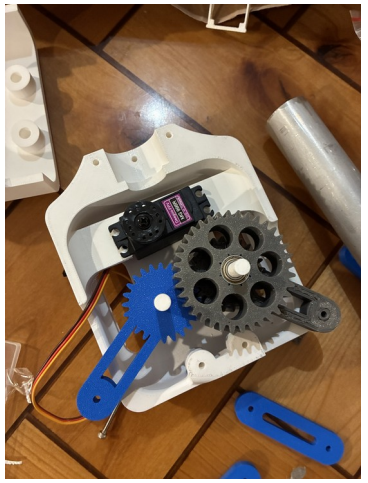
Source: RoomCleaner engineering calculations, “Floor board” (calculated)

# Geared Four-Bar Claw









v6 clamp, final four-bar (CAD), printed claw body and gear train

- Parallel jaws pinch thin fabric; 53 mm opening
- Camera moved into the claw to see each grasp
- Stall check moved the gears from module 1 to 1.5

**Gear stress at servo stall**

$\sigma = F_t / (b\ m\ Y) =$ **11.0 MPa**

$SF = 22.5\ MPa / 11.0\ MPa =$ **2.05**

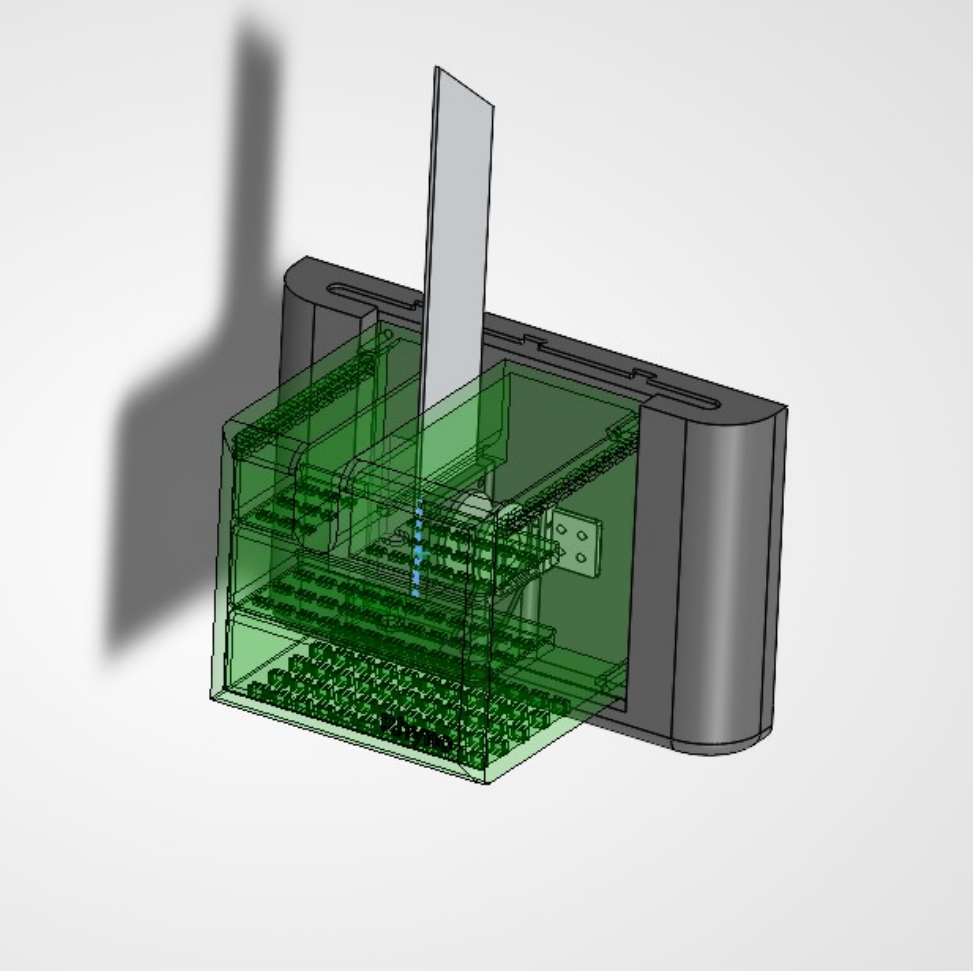
Phase 4

# Final Design

October 7 – October 21

Size the winch, the cooling and the weight in SolidWorks

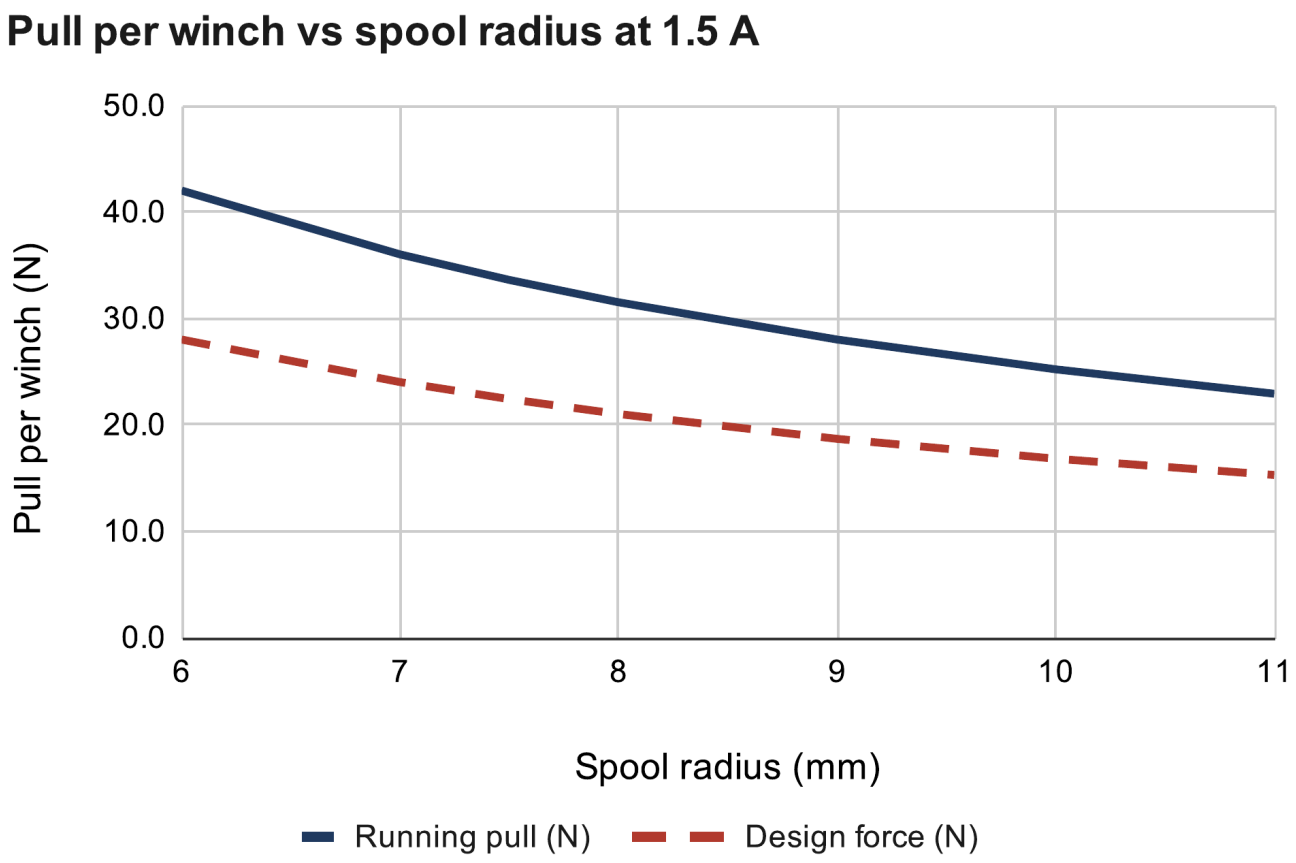
# Final Wall Station



Final wall station (SolidWorks), shell shown transparent

- Slide-in wall mount: the station lifts off while the anchor stays in the wall
- Vented shell around the motor, spool, encoder and controller
- Aluminum fin leaves through a slot in the shell top
- Three station and three spool versions in SolidWorks

# Winch Pull at 1.5 A




Source: RoomCleaner engineering calculations, “Winch pull” (calculated)

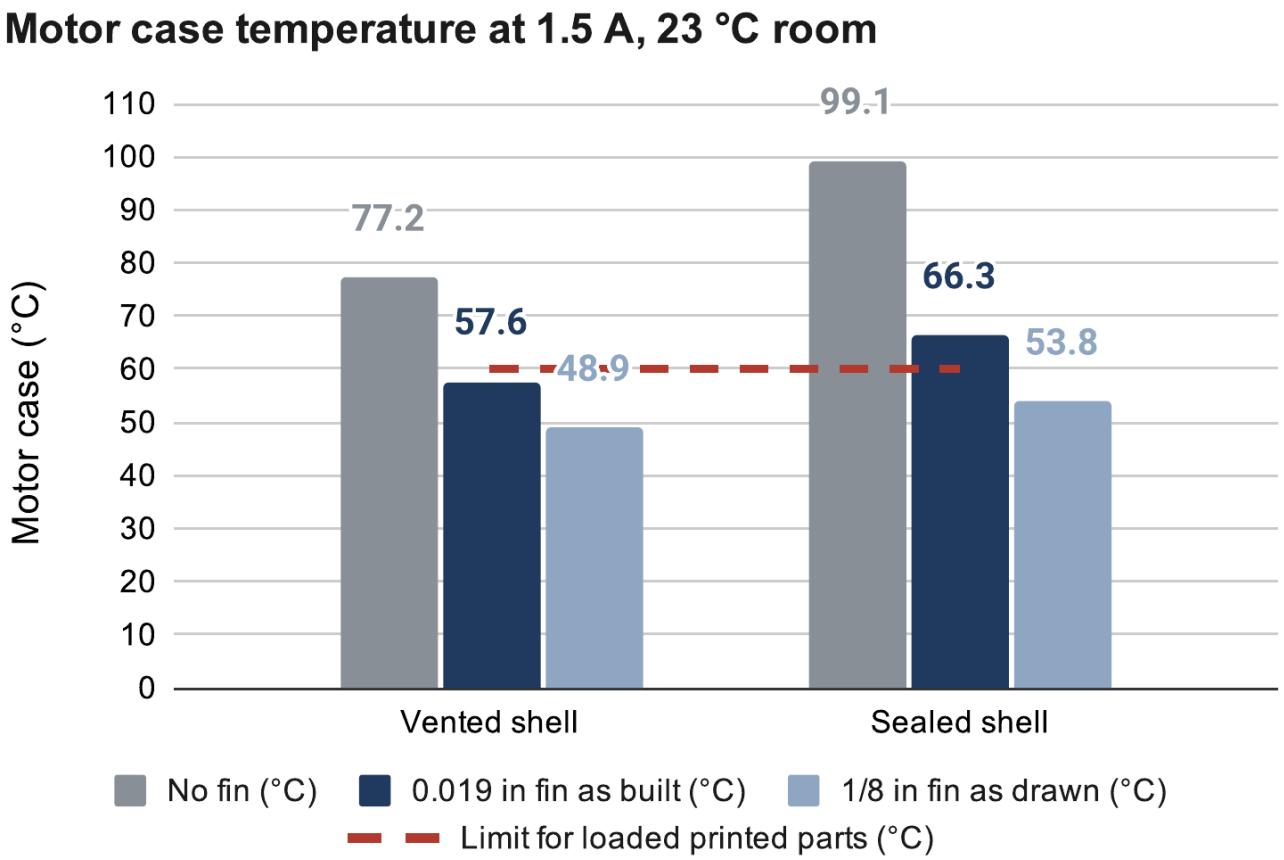
- NEMA 17 (National Electrical Manufacturers Association) stepper on a printed spool
- Driver current raised from 1.06 A to the rated 1.5 A: 41 % more pull
- 34 N with an empty spool, 25 N when full

**Winch design force**

$T = \tau / r = 0.252\ \text{N·m} / 0.010\ \text{m} = \mathbf{25.2\ N}$

$F_{design} = 25.2\ \text{N} / 1.5 = \mathbf{16.8\ N}$

# Motor Cooling




Source: RoomCleaner engineering calculations, “Motor temperature” (calculated)

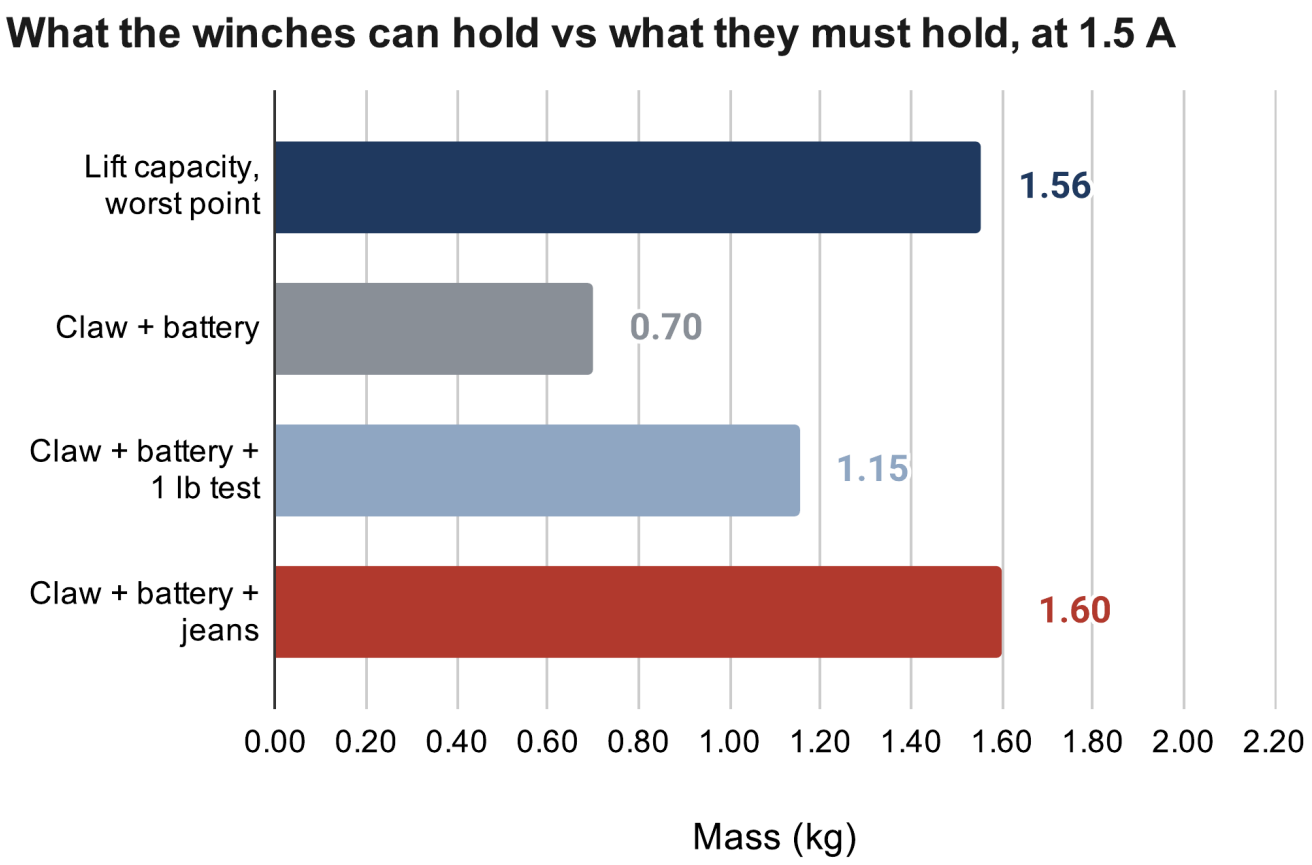
- 1.5 A doubles motor heat to 4.05 W
- PETG creeps above 60 °C, so the shell is vented and finned
- Motor at 45–49 °C with the 1/8 in fin, 56–59 °C with the 0.019 in prototype

**Motor heat, $P = I^2 R$**

At 1.06 A: $(1.06\ \text{A})^2 \times 1.8\ \Omega = 2.0\ \text{W}$

At 1.5 A: $(1.5\ \text{A})^2 \times 1.8\ \Omega = \mathbf{4.05\ W}$

# Weight Budget: Open Issue




Source: RoomCleaner engineering calculations, “Lift budget” (calculated)

- Lift capacity at 1.5 A: 1.56 kg
- Claw weighed at 0.60 kg; the CAD estimate was 0.76 kg
- 1 lb test fits; jeans are 0.04 kg over
- Next: the pull test, then a 40 g cut or gears at each winch

**Lift capacity**

$m_{max} = F_{design} / (k\,g)$

$= 16.8 / (1.101 \times 9.81) =$ **1.56 kg**

Needed: $0.60 + 0.10 + 0.90 =$ **1.60 kg**

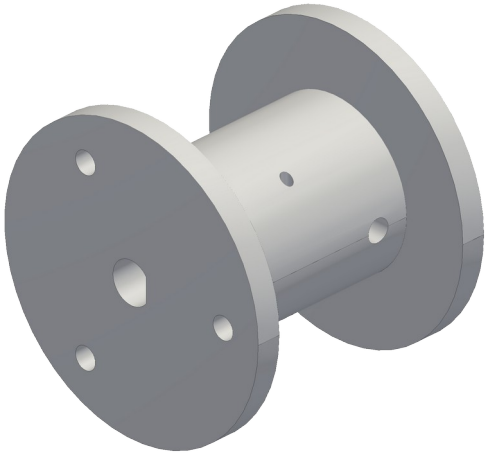
Phase 5

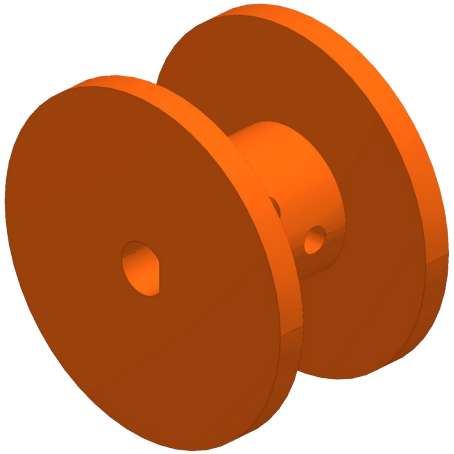
# Build and Test

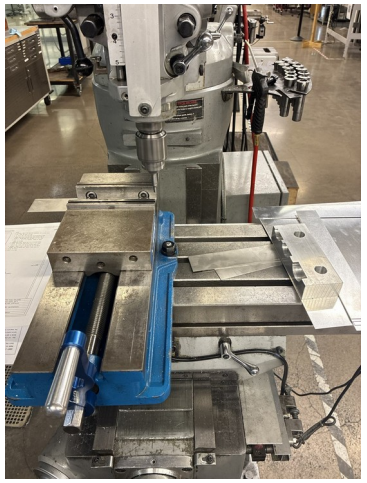
October 22 – October 27

Build the stations and run the first system test

# Printing and Machining





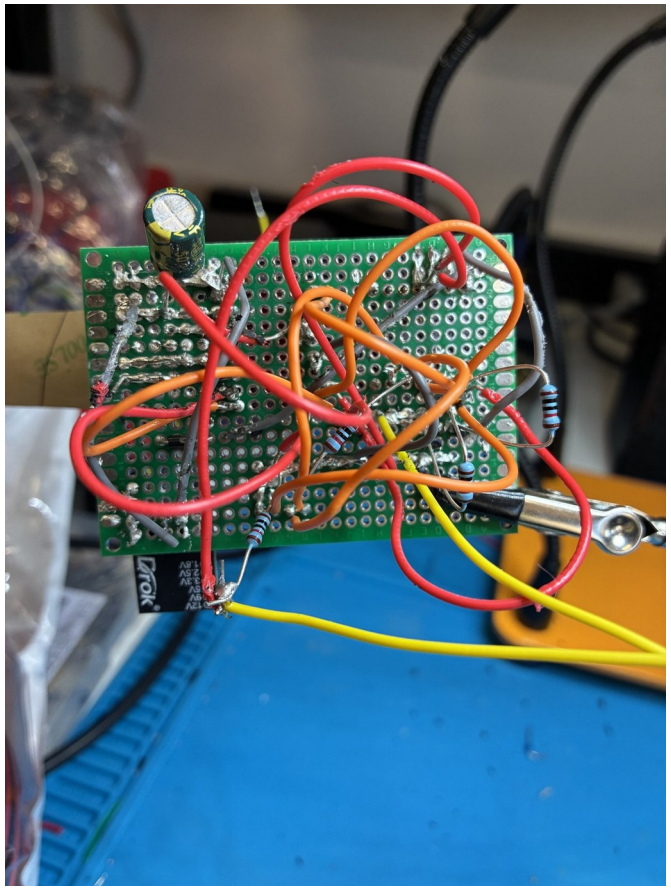




Spool v2 (bore too tight) and v3, fin mill setup, finished fin

- Printed holes come out at least 0.2 mm undersize: the spool took three prints
- Fit coupons are now printed before full parts
- Fin holes drilled in one mill setup to ±0.1 mm

# Electronics Bring-Up





- Driver stays off at boot, so a reset cannot move a motor
- First motor failed to turn: logic rail at 1.14 V, not 3.3 V
- Supply rails are now checked with a meter before any code

Soldered station board and bench bring-up

# First System Test

*[Photo or video from the three-station test]*

| Check | Pass limit | Result |
|---|---|---|
| **Self-check** | All stations answer; slack taken up | *[fill in]* |
| **Calibration** | Held-out poses inside the margin | *[fill in]* |
| **Short move, 1 lb** | 300 mm at 50 mm/s, ≤ 2 mm per cable | *[fill in]* |
| **Pickup** | One garment into the hamper | *[fill in]* |

- Three stations: the fewest cables that fully hold the claw
- Self-check, calibration, short move, then pickup
- 1 lb first, then one garment; pass limits set before the run

# Thermal Verification

*[SolidWorks temperature plot: motor, fin, screws and mount]*

| Method | 0.019 in fin | 1/8 in fin |
|---|---|---|
| **Hand calculation** | 56–59 °C | 45–49 °C |
| **SolidWorks** | *[fill in]* | *[fill in]* |
| **Hold test** | *[fill in]* | *[fill in]* |

- PETG mount must stay below 60 °C
- Hand calculation, SolidWorks study, then a 90-minute hold test
- The PETG runs at about the motor temperature

# Results Against Targets

| Target | Goal | Current | Status |
|---|---|---|---|
| **Lift capacity** | 1.60 kg with jeans | 1.56 kg (calculated) | **0.04 kg short** |
| **Claw mass** | 0.45 kg | 0.60 kg (weighed) | **Not met** |
| **Motor temperature** | Below 60 °C | 45–49 °C with the 1/8 in fin | Met in design |
| **Working area** | Full 11 × 9 ft floor | All but 0.2 m along the walls | Met in design |
| **Wiring** | One cable per station | One 12 V pair; 0.51 V drop | Met |
| **Accuracy** | 2 mm per cable | 0.7 mm (calculated) | To be tested |
| **Speed** | 50 mm/s at 2 m/s² | Not yet run | To be tested |
| **Cost** | Hobby budget | *[total cost]* | *[fill in]* |

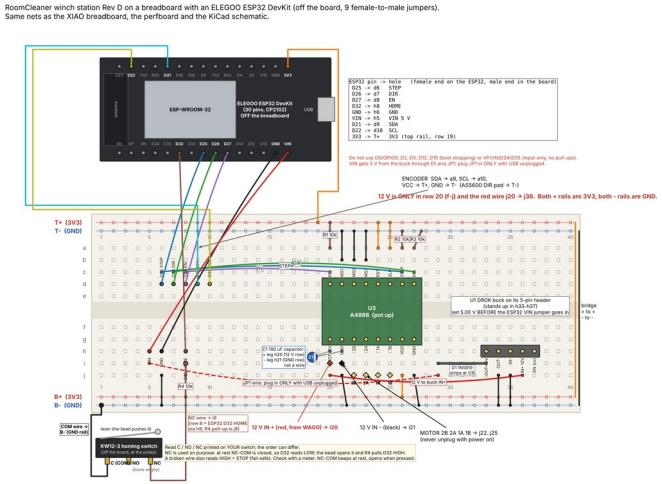
# Good and Poor Decisions

| Good decision | Result |
|---|---|
| **One controller per winch** | Five times less wire; each station tested alone |
| **Camera calibration instead of bead homing** | Whole room reachable |
| **Gear strength checked at servo stall** | Failure caught in CAD, before printing |

| Poor decision | Result |
|---|---|
| **Mass budget set late** | Claw 0.15 kg over budget; jeans 0.04 kg over capacity |
| **Bead homing as the first plan** | Under a third of the room reachable |
| **Several stations built before one was proven** | Every fix repeated on each station *[confirm]* |

# Thorough vs. Scrappy

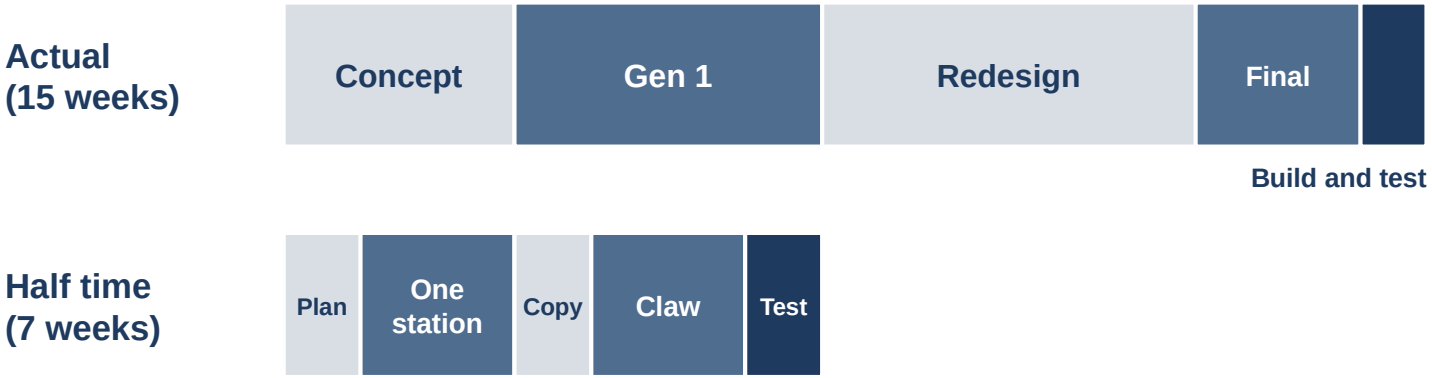





- Thorough where a mistake is expensive or unsafe
  - Heat fin: one mill setup, ±0.1 mm
  - Gears: strength checked before printing
- Scrappy where it is cheap to undo
  - Failed controller replaced by a jumpered ESP32 board
  - Floor board printed on paper

Thorough: fin mill setup. Scrappy: jumpered ESP32 controller board.

# Takeaways and Next Steps




Same time scale. Half-time plan: prove one station, then copy it.

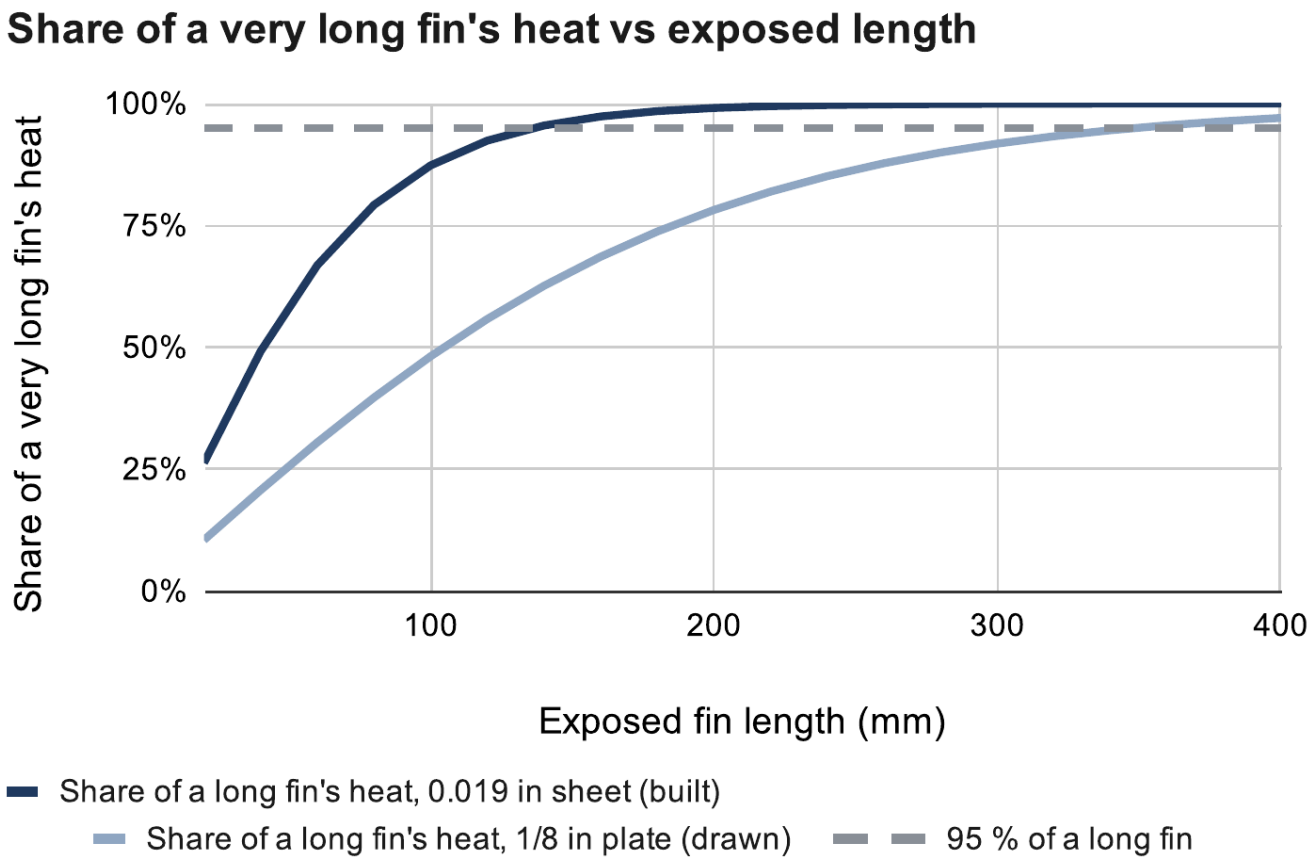
- Takeaways
  - Set the mass budget first, from the measured winch pull
  - Prove one station before building the rest
  - In half the time: freeze requirements in week 1, order long-lead parts on day 1
- Next steps
  - Pull test; then a 40 g claw cut or gears at each winch
  - 90-minute hold test at 1.5 A with the 1/8 in fin
  - Three-station test, then add the fourth station

# Questions

Jasper Buntinx | jasper_buntinx@tamu.edu

# Appendix

# Heat Fin Length




- 0.019 in prototype reaches 95 % of its capacity by 136 mm
- 1/8 in design keeps gaining to 349 mm
- Built at 200 mm, limited by ceiling clearance

**Fin length and efficiency**

$m = \sqrt{(2h / (k\,t))}, \quad L_{95} = 1.83 / m$

$\eta = \tanh(mL) / (mL) = \mathbf{0.74}$ (1/8 in)

Source: RoomCleaner engineering calculations, “Heat fin” (calculated)

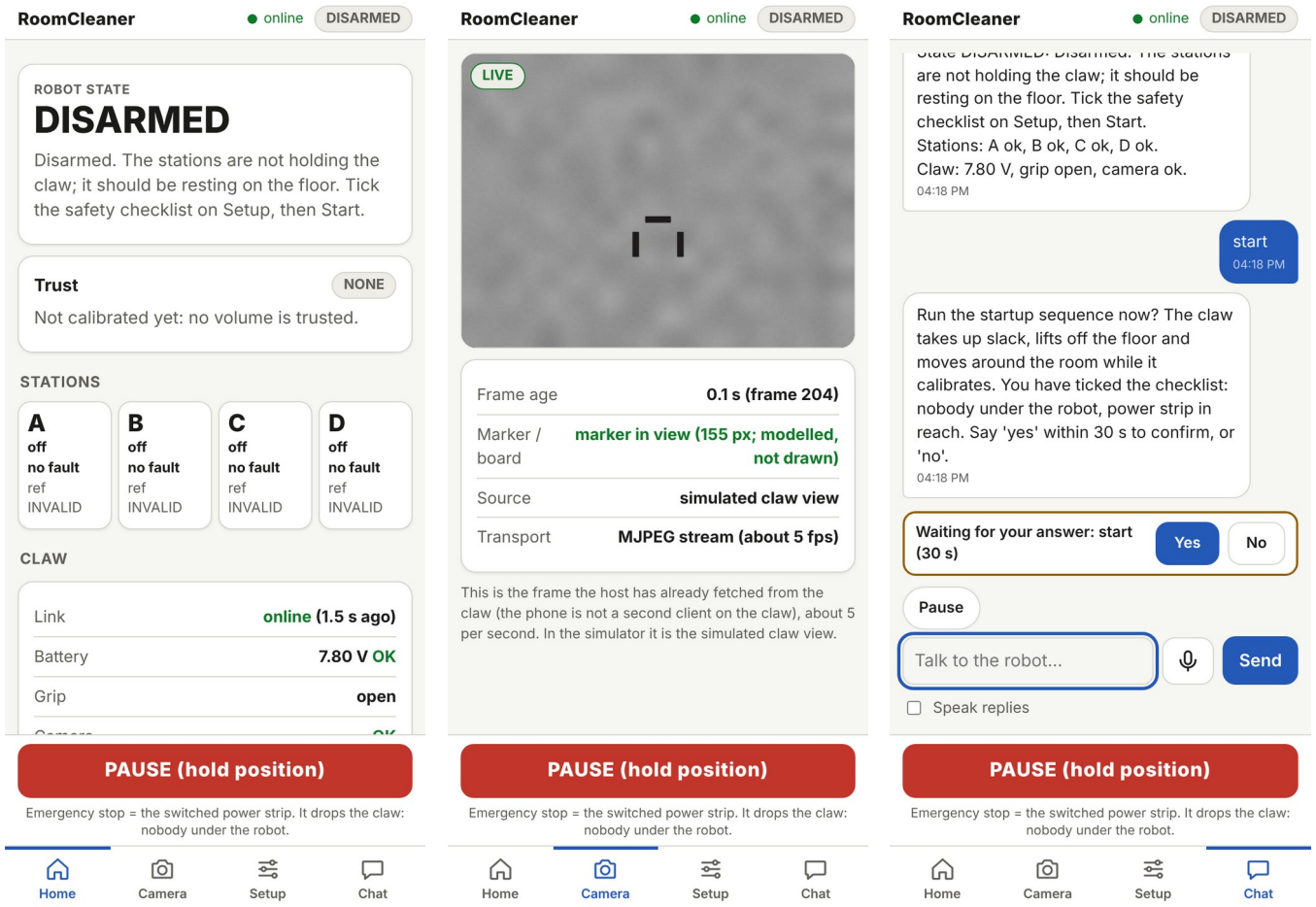
# Fin Sizing for 4 W

| Painted 6061, 44 mm wide | Mass | Shortest for ≤ 60 °C |
|---|---|---|
| **0.019 in (prototype)** | 14 g | None (2.0 W max) |
| **1/16 in** | 47 g | None (3.6 W max) |
| **0.090 in** | 68 g | 210 mm or none |
| **1/8 in (design)** | 94 g | 145–185 mm |
| **3/16 in** | 141 g | 125–140 mm |
| **1/4 in** | 189 g | 115–125 mm |

All 4.05 W through the fin, 23 °C room, paste contact. Calculated.

- If the fin alone had to carry all 4.05 W
- 0.019 in sheet: not possible at any length
- 1/8 in plate: 145–185 mm

# Operating Sequence



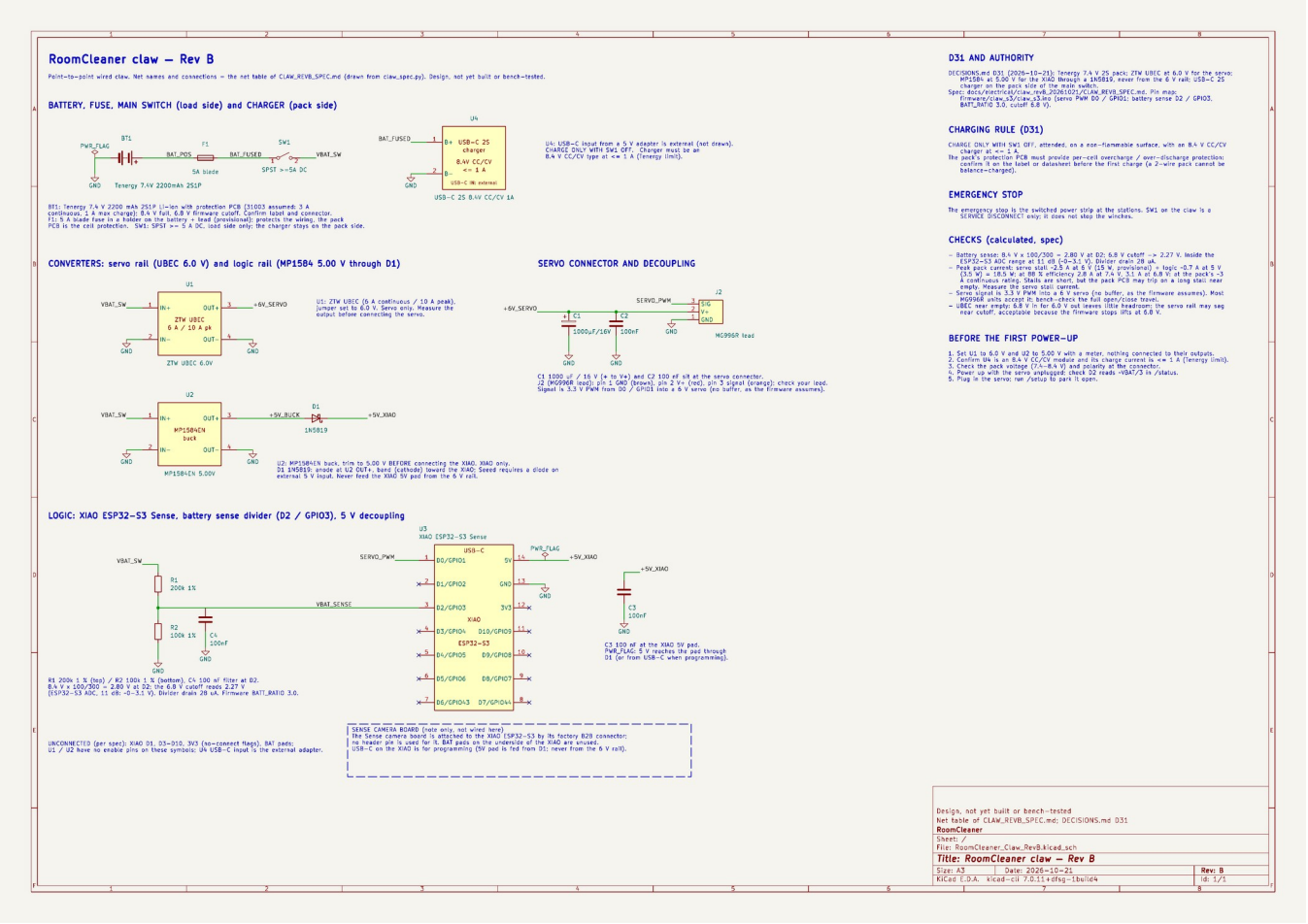

Phone app: home, live claw camera and chat

- Start up: self-check, take up slack, calibrate
- Find: scan from above, detect garments, choose the nearest
- Pick: hover until still, grip, deliver; the camera confirms each step
- Every move waits for approval on the phone; PAUSE holds position

# Control Logic: Generation 1 vs. Final

| | Generation 1 | Final |
|---|---|---|
| **Where is the claw?** | Assumed from the commanded cable lengths | Measured by the claw camera; refit at every startup |
| **Room geometry** | Anchors from a tape measure | Anchors and cable lengths fitted, with error bars |
| **Motor control** | One central board | A controller per station, coordinated by the laptop |
| **Finding laundry** | Ceiling camera | Claw camera |
| **Did the grab work?** | Assumed | Camera checks pickup and delivery |
| **Faults** | Emergency stop only | A stop keeps torque; nothing restarts by itself |

# Claw Electrical



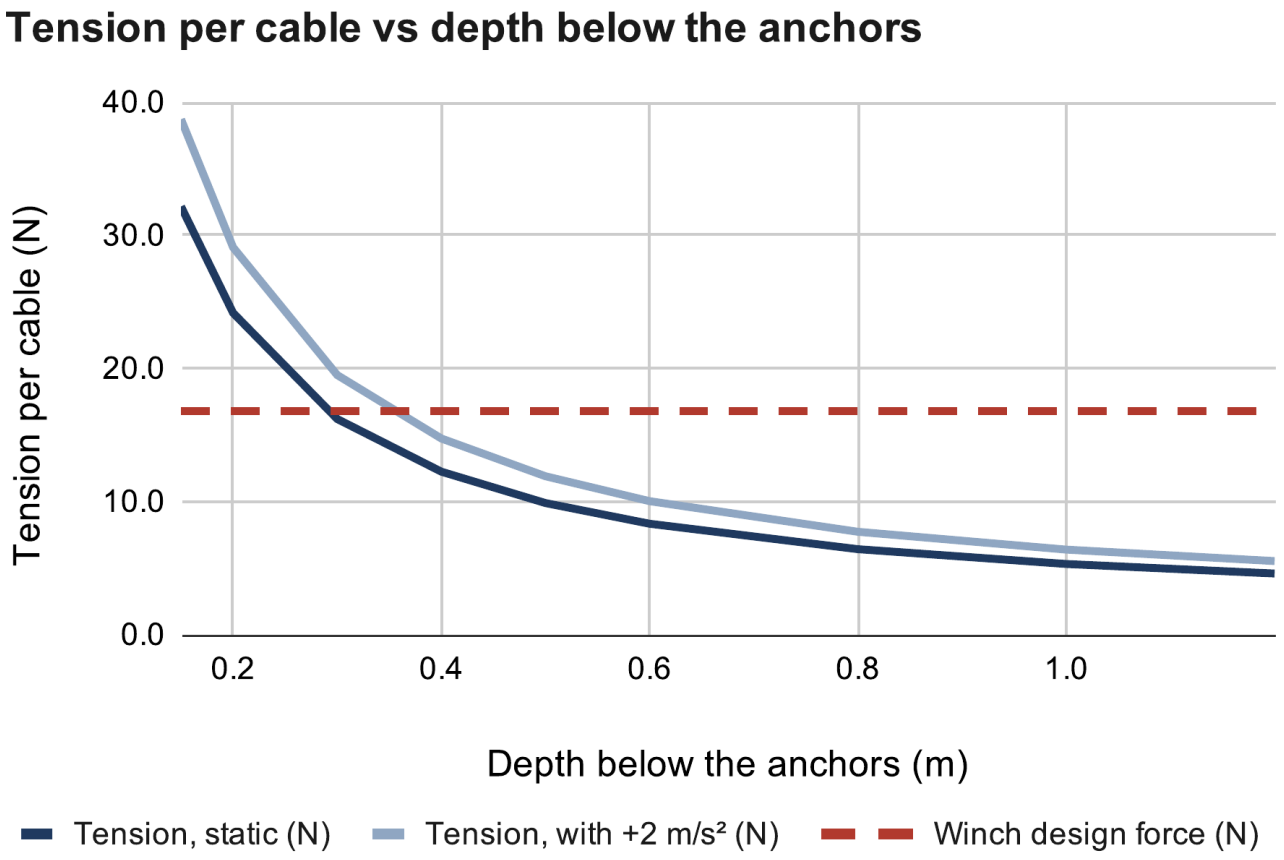

Claw schematic

- Two-cell lithium-ion pack: 7.4 V with a 5 A fuse
- 6 V servo rail and a separate 5 V logic rail
- Peak 2.8–3.1 A against a 3 A pack rating

# Acronyms

| Term | Meaning |
|---|---|
| **CAD** | Computer-aided design |
| **PETG** | Polyethylene terephthalate glycol (the printed plastic) |
| **TPU** | Thermoplastic polyurethane (flexible printed plastic) |
| **NEMA 17** | National Electrical Manufacturers Association motor frame, 1.7 in square |
| **CNC** | Computer numerical control |
| **ESP32** | A Wi-Fi microcontroller family (Espressif) |
| **A4988** | Stepper-motor driver chip (Allegro) |
| **YOLO** | “You Only Look Once”: an object-detection model family |
| **SF** | Safety factor: allowable stress divided by actual stress |

# Calculations and Sources




Source: RoomCleaner engineering calculations, “Cable tension” (calculated)

- All charts come from my engineering calculation sheet
- Incropera et al., Fundamentals of Heat and Mass Transfer, ch. 3
- Allegro MicroSystems, A4988 datasheet

RoomCleaner engineering calculations (Google Sheets)